# 1. Product Introduction

## 1.1. Product Description

Mini2 Series of uncooled infrared modules, Contains 640 × 512/384 × 288/256 × 192 resolution, and the entire series uses self-developed high refresh rate 12um WLP package detector, It has the characteristics of high performance, small size, light weight, low power consumption and low cost , meeting SWaP (Size, Weight and Power/Price) application requirements.

The entire series is equipped with an unobstructed algorithm, which is smooth and lag-free to use. It is equipped with a new generation of infrared -specific image algorithm , and the image quality is delicate and clear. The module provides a variety of digital interfaces and can be flexibly connected to various intelligent processing platforms.

## 1.2. Product Features

- High refresh rate across the board
- Standard no-block algorithm
- High-quality images
- High cost performance



# 2. Product Selection

## 2.1. Model Coding Rules

Table 2-1 Mini2 module model coding rules

| Mini2384 | 09110X | H | NR | M | A |
|---|---|---|---|---|---|
| Product Name | Lens (focal length, lens F number) | Gain Mode | Movement mode | Lens focus type | Lens serial number |
| Mini2256<br>Mini2384<br>Mini2640 | 00000X: No lens<br>09110X: 9.1mm F1.0<br>18011X: 18mm F1.1 55010X: 55mm F1.0 For more lens codes, please refer to See "2.3 Lens Parameters number" | H: High image quality | NR: Imaging | S: Fixed focus<br>M: Manual adjustment<br>N: No lens | A: The focal length, f number, The first focus mode Lenses<br>B: The second lens |

Note 1: Supports lens-free selection.

## 2.2. Available models

Table 2-2 Mini2 module optional models

| Modules / FOV Selection / Lenses | | Mini2 256 | Mini2 384 | Mini2 640 |
|---|---|---|---|---|
| No lens | | × | × | × |
| | | Mini225600000XHNRNA | Mini238400000XHNRNA | Mini264000000XHNRNA |
| 3.9mm F1.1 Fixed focus | | 43.3°×32.8° | 65°×50° | × |
| | | Mini225603911XHNRSA | Mini238403911XHNRSA | × |
| 6.8mm F1.1 Fixed focus | | 26.1°×19.6° | 39.4°×29.4° | × |
| | | Mini225606811XHNRSA | Mini238406811XHNRSA | × |
| 9.1mm F1.0 Manual adjustment | | 19.4°×14.6° | 29.2°×21.7° | 48.7°×38.6° |
| | | Mini225609110XHNRMA | Mini238409110XHNRMA | Mini264009110XHNRMA |
| 13mm F1.0 Manual adjustment | | 13.6°×10.2° | × | × |
| | | Mini225601310XHNRMA | × | × |
| 13.5mm F1.0 Manual adjustment | | 13.0°×9.7° | 19.6°×14.7° | 31.9°×25.7° |

Mini2 Series Product Brochure

| | Mini225613510XHNRMA | Mini238413510XHNRMA | Mini264013510XHNRMA |
|---|---|---|---|
| 15mm F1.0 Manual adjustment | 11.7°×8.8° | 17.5°×13.1° | × |
| | Mini225615010XHNRMA | Mini238415010XHNRMA | × |
| 18mm F1.1 Manual adjustment | 9.8°×7.3° | 14.7°×11.0° | 24.2°×19.5° |
| | Mini225618011XHNRMA | Mini238418011XHNRMA | Mini264018011XHNRMA |
| 25mm F1.0 Fixed focus | 7.0°×5.3° | 10.5°×7.9° | 17.5°×14.0° |
| | Mini225625010XHNRMA | Mini238425010XHNRMA | Mini264025010XHNRMA |
| 35mm F1.0 Fixed focus | 5.0°×3.8° | 7.5°×5.7° | 12.5°×10.0° |
| | Mini225635010XHNRMA | Mini238435010XHNRMA | Mini264035010XHNRMA |
| 55mm F1.0 Fixed focus | 3.2°×2.4° | 4.8°×3.6° | 8.0°×6.4° |
| | Mini225655010XHNRMA | Mini238455010XHNRMA | Mini264055010XHNRMA |



Note 1: All the above lenses are of type A. If there are any updates later, the annotations will be updated.

Note 2: Above marked with “×”, which means that this option is not available.

Note 3: If you need a combination other than the ones in the recommended list, you can contact us for customization.

## 2.3. Lens parameters

Table 2-3 Lens parameters

| Lenses Code | Lens parameters | | | Lens Depth of Field | Detection distance | Identification distance | Identification distance | weight (Lens + Flange) |
|---|---|---|---|---|---|---|---|---|
| | focal length mm | F-number | IFOV | | | | | |
| 3D911X | 3.9 | 1.1 | 3.07mrad | 0.3m~∞ | 0.43km | 0.1km | 0.054km | 10g±2% |
| 06812X | 6.8 | 1.2 | 1.76mrad | 0.8m~∞ | 0.76km | 0.19km | 0.09km | 12g±5% |
| 09110X | 9.1 | 1.0 | 1.3mrad | 1.73m~∞ | 1.1km | 0.25km | 0.17km | 10.5g±5% |
| 13010X | 13 | 1.0 | 0.92mrad | 3.52m~∞ | 1.4km | 0.36km | 0.18km | 13.2g±5% |
| 13510X | 13.5 | 1.0 | 0.9mrad | 3.8m~∞ | 1.5km | 0.37km | 0.19km | 15.2g±5% |
| 15010X | 15 | 1.0 | 0.8mrad | 4.7m~∞ | 1.7km | 0.42km | 0.21km | 20.48g±5% |
| 18011X | 18 | 1.1 | 0.7mrad | 6.14m~∞ | 2km | 0.5km | 0.25km | 25g±5% |
| 25010X | 25 | 1.0 | 0.48mrad | 13.02m~∞ | 2.8km | 0.7km | 0.35km | 67g±2% |
| 35010X | 35 | 1.0 | 0.34mrad | 25.52m~∞ | 3.9km | 0.97km | 0.49km | 111g±2% |
| 55010X | 55 | 1.0 | 0.22mrad | 63m~∞ | 6.1km | 1.5km | 0.76km | 191g±2% |

Note 1: The above detection distance, recognition distance and identification distance are estimated based on the Johnson criterion, taking pedestrians (1.8×0.5×0.3m) as the target.

Note 2: The weight here refers only to the lens weight. The full weight includes the lens , module body, connecting cables, and optional USB components.

Note 3: This table does not include the field of view (FOV) data. For this data, please refer to “2.2 Optional Models” in this document.

Note 4: The depth of field of the lens refers to the maximum clear range that can be obtained when the focus is fixed (dispensing or fixing with a stop screw) , non-focusing range, Focus range

The model value is 0.3m~∞.

# 3. Parameter Introduction

Table 3-1 Mini2 module product performance parameters

| Overview | MINI2 | | |
|---|---|---|---|
| Detector Type | Vanadium Oxide Uncooled Infrared Focal Plane Detector | | |
| Resolution | 256×192 | 384×288 | 640×512 |
| Detector frame rate | 25/50Hz | 30/60Hz | 30/60Hz |
| Pixel spacing | 12μm | | |
| Response band | 8～14μm | | |
| Noise Equivalent Temperature Difference NETD | ≤40mK@25℃,F#1.0,25Hz | | |
| Thermal time constant | ＜12ms | | |
| Image Adjustment | | | |
| Non-uniformity correction | Shutter correction/no-blocking algorithm correction | | |
| Brightness/contrast adjustment | 0～100 levels are optional | | |
| Polarity/false color | White hot/black hot/support multiple pseudo colors | | |
| Electronic zoom | 1.0～8.0× continuous zoom (step length 0.1) | | |
| Image Mirroring | Up and down/left and right/diagonal | | |
| Electrical parameters | | | |
| Simulation Video | PAL/NTSC | | |
| Digital Video | DVP/BT656/USB2.0/MIPI | | |
| Communication interface | UART/I2C/USB2.0 | | |
| Power input | 5V | | |
| USB output typical power consumption | ＜0.35W | ＜0.42W | ＜0.7W |
| Physical properties | | | |
| Weight (without lens) | <7g | | <8.6g |
| Dimensions (without lens) | 21×21×10.3mm | | |

| Environmental adaptability | |
|---|---|
| Operating temperature | -40℃~+80℃ |
| Storage temperature | -50℃~+85℃ |
| wet Spend | 5%-95%, No condensation |
| Vibration move | 6.06g, random vibration, all axes |
| rush hit | 80g @4ms, The rear peak sawtooth wave, 3 Axis 6 Towards |

Note 1: The above parameters are all measured under laboratory conditions. The actual specifications and test methods are subject to the test report.

# 4. Module interface and expansion components

## 4.1. Module interface description

50PIN male connector model: DF40C-50DP-0.4V (51)

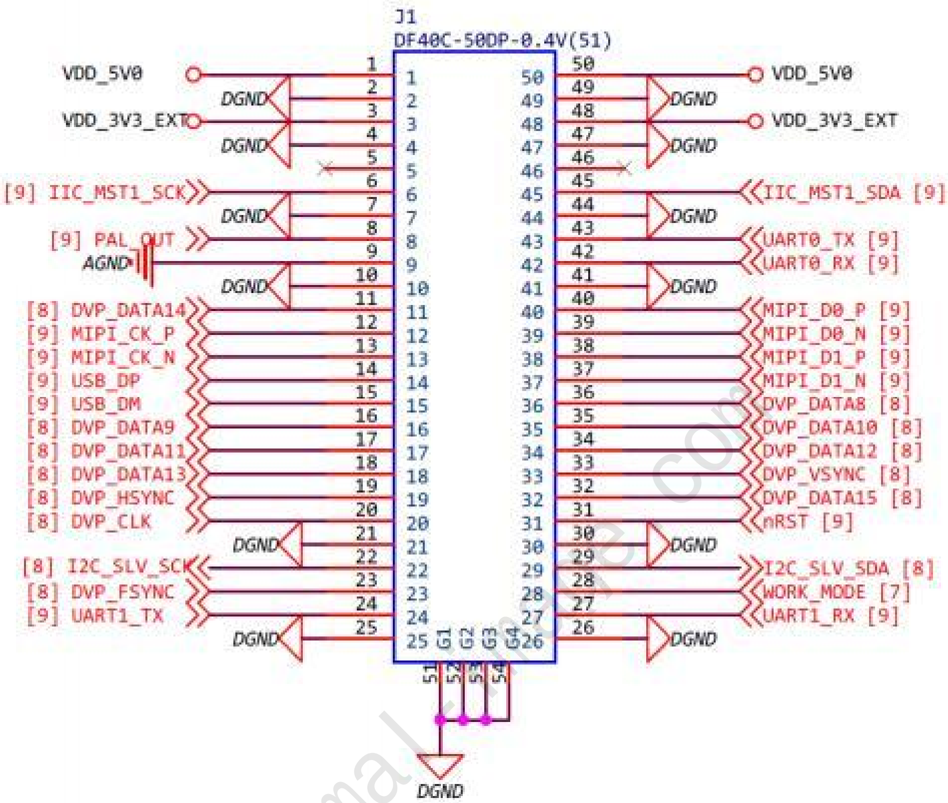



Figure 4-1 Module product pin diagram A

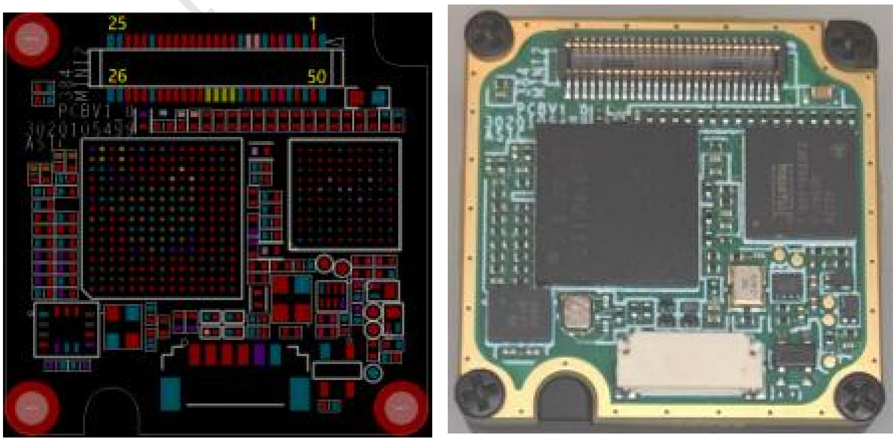



Figure 4-2 Module product pin diagram B

Table 4-1 HIROSE50 Core Connector User Interface Definition

| Pin Number | Pin Name | type | illustrate |
|---|---|---|---|
| 20. 33. 19.<br>32. 11. 18. 34.<br>17. 35. 16. 36 | DVP_CLK<br>DVP_VSYNC<br>DVP_HSYNC<br>DVP_DATA15~8 | Output | DVP data signal DATA15:MSB DATA8:LSB |
| 12. 13. 40. 39.<br>38. 37 | MIPI_CK_P/N<br>MIPI_D0_P/N<br>MIPI_D1_P/N | Output | 2lane MIPI data signal |
| 14. 15 | USB_DP USB_DM | Output | USB2.0 data signal |
| 8. 9 | PAL_OUT AGND | Output | Analog output signal Analog output ground |
| 43. 42 | UART0_TX<br>UART0_RX | Output | Universal communication serial port pins, Can be used to control modules |
| twenty four, 27 | UART1_TX<br>UART1_RX | I/O | The serial port's transmit and receive pins for debugging. Printable module running logo Can also be used as normal I/O |
| twenty two, 29 | I2C_SLV_SCK<br>I2C_SLV_SDA | I/O | IIC slave The clock pin IIC slave Data pin |
| 6. 45 | IIC_MST1_SCK<br>IIC_MST1_SDA | I/O | IIC master The clock pin IIC master Data pin |
| twenty three | DVP_FSYNC | I/O | The module's external synchronization input signal, Synchronous signal output by non-DVP It can also be used as a normal I/O |
| 28 | WORK_MODE | Input | Set high to enter update mode. The hardware design can be left floating when not in use |
| 31 | Nrst | Input | Set low to enter reset mode, the hardware design can be left floating |
| 1. 50 | VDD5.0 | Power | External power input 5V, the module can work normally with a single 5V power supply do |
| 3. 48 | VDD3.3 | Power | External power input 3.3V, only reserved, No power supply required |
| 2. 4. 7. 10. twenty one, | DGND | Power | GROUND |

| | | | |
|---|---|---|---|
| 25. 26. 30. 41.<br>44. 47. 49 | | | |
| 5. 46 | NC | NC | User unavailable |

Note 1: The order of pin numbers in the table is consistent with the order of pin names .

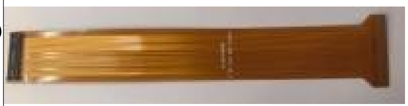
Note 2: DVP, IIC slave Voltage default 1.8V, can be switched by command 3.3V.

Note 3: UART0, UART1, I IC master The voltage is fixed at 3.3V. Unable to switch.

Note 4: The hardware description here is for selection reference only. Please contact technical personnel to obtain detailed hardware development information during specific development .

Note 5: CVBS only supports black and white hot switching.

## 4.2. Module expansion components and cables

| type | Product Legend | Material Code | Main interfaces/functions |
|---|---|---|---|
| Cable: 50pin to 50pin |  | 2030300095 | 50pin full signal extension cable |
| Cable: 50pin to Goldfinger |  | 2030300199 | 50pin full signal to gold finger cable |
| USB Board |  | 2030110669 | USB2.0 Drawing, model Simulation See Frequent Diagram, serial communication, USB power supply |
| Type-c data cable-CVBS |  | 3030300204 | Analog video output, serial communication |

# 5. Structural dimensions

Note: Here we only take mini2 384 without lens module and mini2 384 9.1mm lens module as examples. picture Please contact our technical staff for more information.

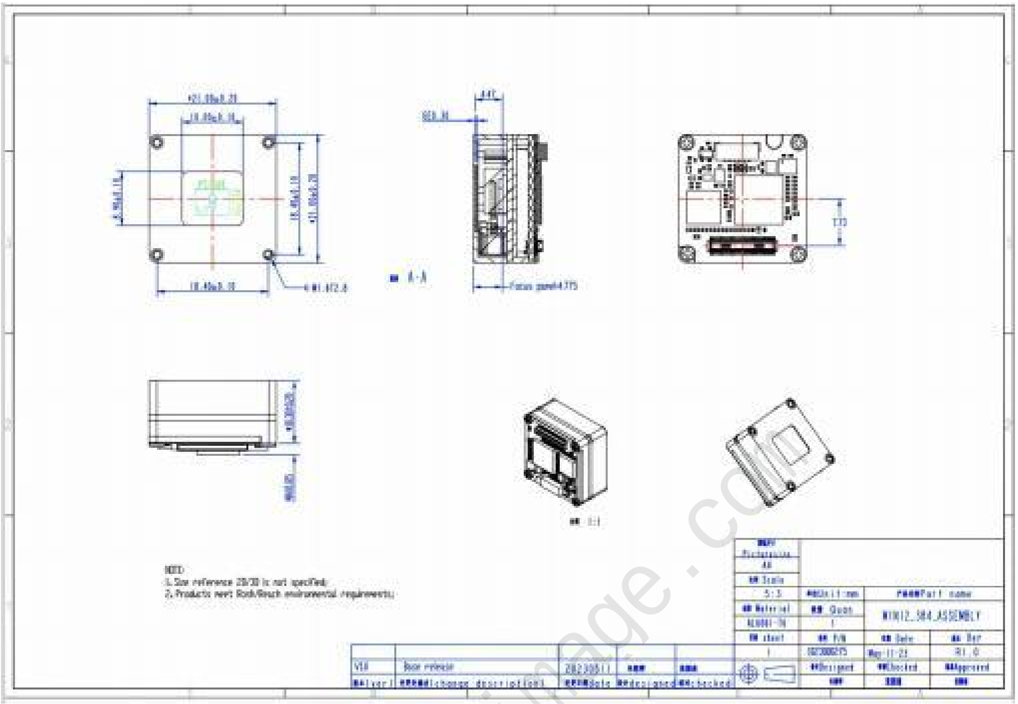


Figure 5-1 Mini2 384 lensless module structure

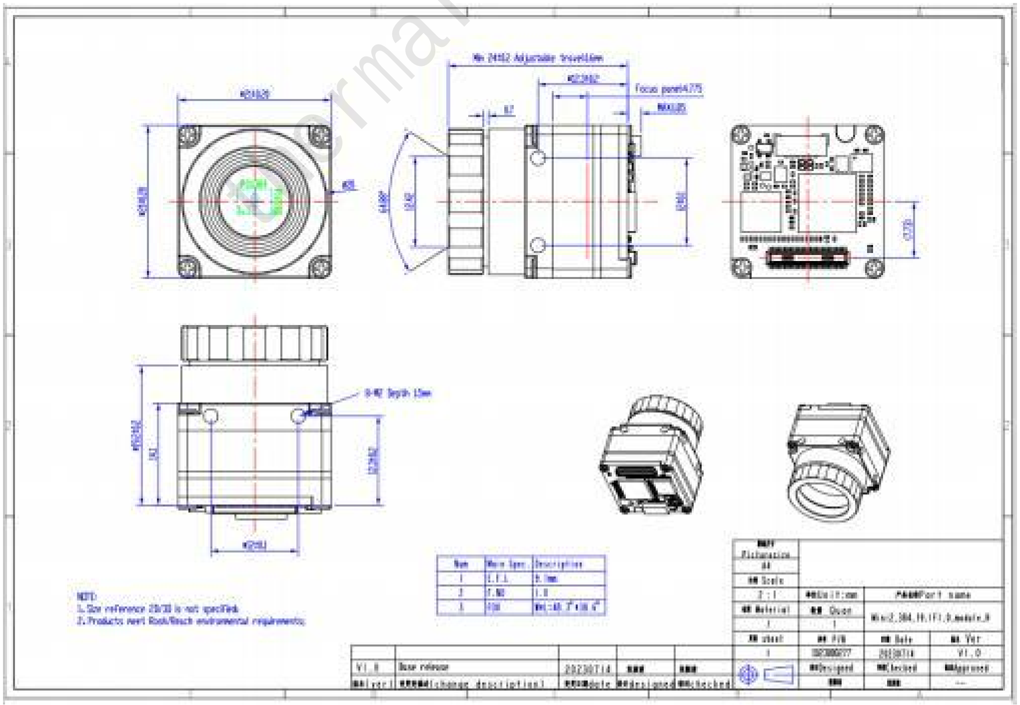


Figure 5-2 Mini2 384 9.1mm lens module structure diagram